

Function

```
def function_name(parameter)

return -(optional)

////////// def greet(x): print("welcome",x)

greet(4)

//////////
```

return values

```
def add(a,b)

    result = a+b
    return result

add(5,4)

//////////
```

```
In [ ]: def greet(x):
        print("welcome",x)

        greet(4)
```

welcome 4

```
//////////
```

An iot temperature sensor reads 5 readings/min write a python code to store these readings in a list print all, write a function to find the average

```
sensor={"sensor1:} sensor={'s1':23.4,'sensor2':15.7,'sensor3':2.51}

//////////
```

```
In [12]: #list tuples dictionary

def pr(sensor):
    print("Sensor values:", sensor)

def avg4(sensor):
    sum=0
    for j in range(5):
```

```

        sum=sum+sensor[j]
    avg=sum/5
    return avg

sensor=[]
for i in range(5):
    a=int(input("Enter sensor value: "))
    sensor.append(a)

print(sensor)
val = avg4(sensor)

print("average value: ",val)

```

[1, 2, 5, 3, 1]
average value: 2.4

```

In [ ]: lst=[]
for i in range(5):
    x=int(input("enter the no.:"))
    lst.append(x)

def display():
    print(lst)

def average(lst):
    result=sum(lst)/len(lst)
    return result

# function calls
display()
print("average of sensor values is:",average(lst))

```

[1, 2, 5, 2, 1]
average of sensor values is: 2.2

```

In [3]: def aver(read):
        sum=0
        for i in range(5):
            sum=sum+sensor[i]
            avg=sum/5
        return avg

read={1,2,3,4,5}
print(read)

val=aver(read)
print(val)

```

{1, 2, 3, 4, 5}
3.2

1. Use dictionary to store
key -> slot no.(s1-s5) value -> status("empty" or "occupied")
2. use tuple to store parking lot information parking-name, no.of slots
3. use loop to display a menu
- 1.park vehicle functions all 2. remove vehicle
- 3.show parking status 4. // summary -----> parking name,total slots, vehicles parked
5. exit
4. Maintain a list to store vehicle no.s that entered

```
In [ ]: #2
parking_info=("Fast Parking",5)

#1
parking_slots={'s1':"empty",'s2':"empty",'s3':"empty",'s4':"empty",'s5':"empty"}
vehicleslist=[]

#3 (1)
def park_vehicle():
    slot=input("Enter slot number(s1 to s5): ")
    vehicle=input("Enter vehicle no: ")
    if slot in parking_slots: # slot value(1-5)
        if parking_slots[slot]=="empty" in parking_slots[slot]=="occupied":
            vehicleslist.append(vehicle)
    else:
        print("Not empty")
    #else:
    # print("not valid")

def remove_vehicle():

    r=input("Enter vehicles number: ")
    vehicleslist.remove(r)

#def show_status():

park_vehicle()
remove_vehicle()
```